# Multiples of 3 on the chessboard

Animath — Stage olympique de Sophia-Antipolis · 2006 · France

A number is written in each cell of an $8 \times 8$ chessboard. An operation consists in choosing a $3 \times 3$ or $4 \times 4$ square in the board and adding 1 to each number of the chosen square. Can one always, with such operations, make all the numbers of the board multiples of 3 ?